$$A = \int_0^{\pi} y\,dx + \left|\int_{\pi}^{2\pi} y\,dx\right| = \int_0^{\pi} \sin x\,dx + \left|\int_{\pi}^{2\pi} \sin x\,dx\right| = \left[-\cos x\right]_0^{\pi} + \left|\left[-\cos x\right]_{\pi}^{2\pi}\right|$$

$$= \left[-\cos\pi + \cos 0\right] + \left|\left[-\cos 2\pi + \cos\pi\right]\right| = 2 + \left|-2\right| = 4\,.$$

Note

If we compute the definite integral $\int_0^{2\pi} \sin x\,dx$, we get

$$\int_0^{2\pi} \sin x\,dx = \left[-\cos x\right]_0^{2\pi} = \left[-\cos 2\pi\right] - \left[-\cos 0\right] = 0\,.$$

So $\int_0^{2\pi} f(x)\,dx$ does not represent the area of the region bounded by the curve $y = \sin x$, $x-$axis, the lines $x = 0$ and $x = 2\pi$.

Example 9.53

Find the area of the region bounded by $x-$axis, the curve $y = \left|\cos x\right|$, the lines $x = 0$ and $x = \pi$.

Solution

The given curve is $y = \begin{cases} \cos x, 0 \le x \le \dfrac{\pi}{2} \\ -\cos x, \dfrac{\pi}{2} \le x \le \pi \end{cases}$

It lies above the $x-$axis. The required area is sketched in Fig. 9.22. So, the required area is given by

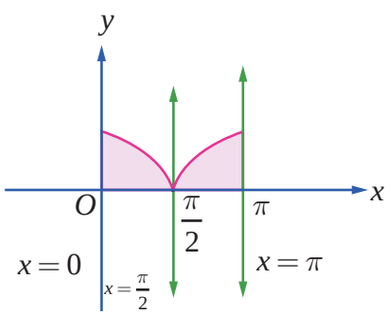


Fig. 9.22

$$A = \int_0^{\pi} y\,dx = \int_0^{\frac{\pi}{2}} \cos x\,dx + \int_{\frac{\pi}{2}}^{\pi} (-\cos x)\,dx = \left[\sin x\right]_0^{\frac{\pi}{2}} - \left[\sin x\right]_{\frac{\pi}{2}}^{\pi}$$

$$= \left[1 - 0\right] - \left[0 - 1\right] = 2\,.$$

9.8.3 Area of the region bounded between two curves

Case (i)

Let $y = f(x)$ and $y = g(x)$ be the equations of two curves in the $XOY-$plane such that $f(x) \ge g(x)$ for all $x \in [a, b]$. We want to find the area A of the region bounded between the two curves, the ordinates $x = a$ and $x = b$.

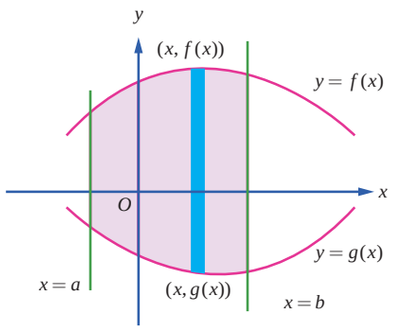


Fig. 9.23

The required area is sketched in Fig. 9.23. To compute A, we divide the region into thin vertical strips of width Δx and height $f(x) - g(x)$. It is important note that $f(x) - g(x) \ge 0$ for all $x \in [a, b]$. As before, the required area is the limit of the sum of the areas of the vertical strips. Hence, we get $A = \int_a^b [f(x) - g(x)]\,dx$.

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